



SPRINT 72

72nd BPSC Prelims Batch

Biology English Notes (Part-1)

BIOLOGY CLASS NOTES

1 - CELL TAXONOMY

1. Microorganisms

Microorganisms are small, typically microscopic organisms, including bacteria, archaea, fungi, algae, protozoa, and viruses. They are classified based on various criteria including **cellular structure, genetics, cell types, and metabolism**. Microorganisms exhibit a variety of differences, including size, structure, habitat, metabolism, and numerous other traits.

Microorganisms can be classified into two types based on their **cell structure**:

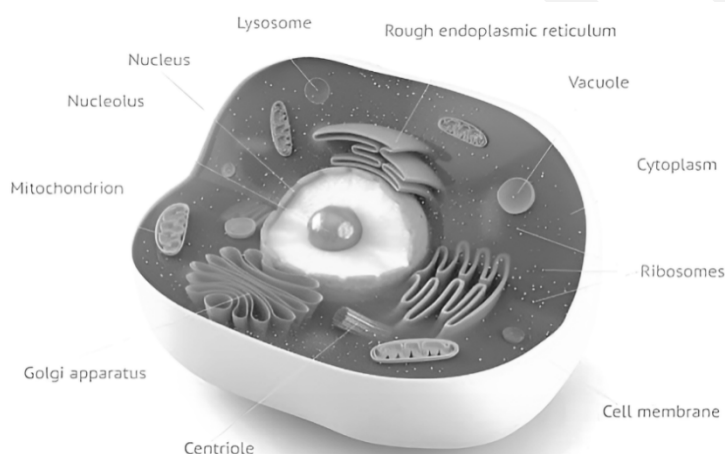
- (i) Acellular
- (ii) Cellular

Acellular – Microorganisms that do not have a cellular structure, such as viruses.

- Do not have cells and **lack a cell membrane**.
- Composed of a few genes protected by a protein coating (e.g., viruses).
- Can live and reproduce only **inside a living cell**.

Cellular – Microorganisms that are composed of one or more cells, such as bacteria, fungi, protozoa, and algae. These are further classified into two types: Prokaryotes and Eukaryotes.

2. Cell



Robert Hooke, while observing cork, described small honeycomb-like compartments and named them “**cells**” (Latin: *small room*). The human body has approximately **37.2 trillion cells**, ranging in size from **3–30 µm** (micrometres in size). A **cell** is the **basic structural, functional, and biological unit of life**.

Unicellular organisms:

- Exist independently.
- Perform all vital life functions.

Thus, the **cell is the fundamental unit of life**.

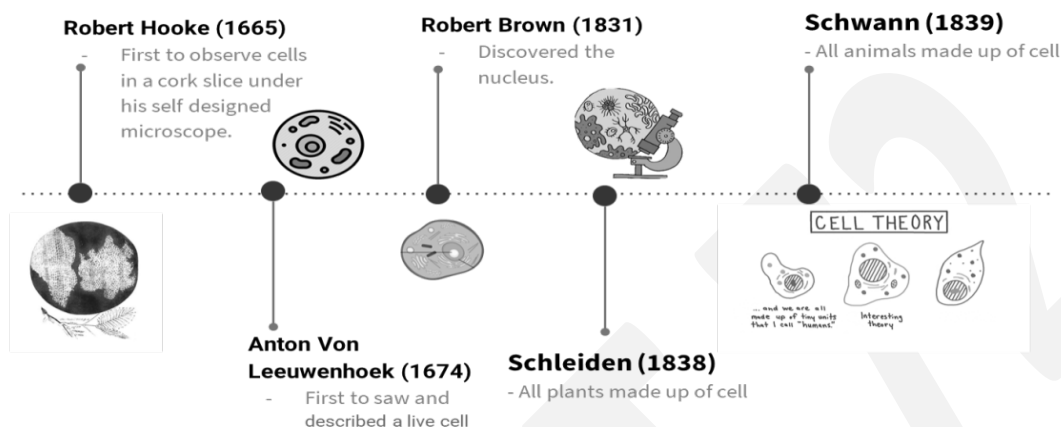
- **Cell Shape and Function:** Cell shape and size depend on the function they perform.
 - Amoeba and Euglena (exhibit traits of both animal and plant cells) → **variable shape**
 - Nerve cells → **fixed, specialized shape**

BCW Fact:

- The smallest cell is *Mycoplasma gallisepticum*, a bacterium measuring 0.1-0.3 µm.

- Longest cell found in the human body is the neuron.
- Largest cell is the egg of Ostrich.
- Largest human cell is the female ovum, whereas the smallest human cell is the sperm cell.

3. History Of Cell Discovery



Evolution of Cell Theory

Matthias Schleiden and **Theodor Schwann** proposed the **cell theory**, but did not explain the origin of new cells.

In 1855, **Rudolf Virchow** added that "**Omnis cellula e cellula**" (all cells arise from pre-existing cells), completing the theory. He modified the original hypothesis proposed by Schleiden and Schwann. The cell theory, as understood today, is:

- All living organisms are **made up of cells and products of cells**.
- A cell is the **fundamental unit** of life.
- All cells originate from **pre-existing cells**.

4. Eukaryotic And Prokaryotic Cells

There are two types of cells - Eukaryotic and Prokaryotic.

- a) **Prokaryotes**: Cells without a nuclear membrane; have no well-defined nucleus; have no membrane-bound organelles (such as endoplasmic reticulum (ER), Golgi apparatus, mitochondria, lysosomes, microbodies, and vacuoles); reproduced by simple fission.

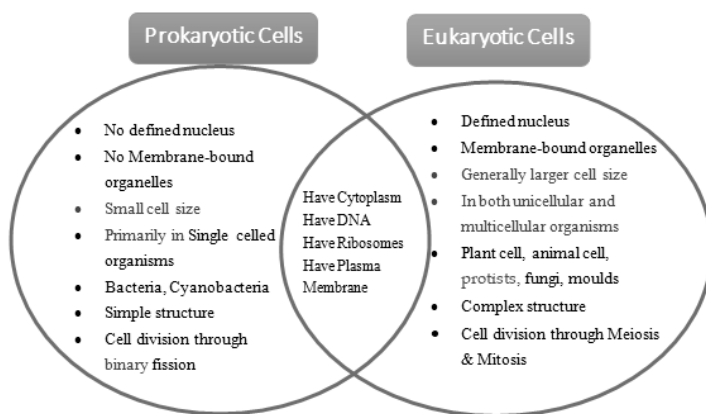
Examples: Bacteria, Blue-green algae (also known as Cyanobacteria, which are capable of photosynthesis), Mycoplasma, and Archaea.

- b) **Eukaryotes**: Cells with membrane-bound Nuclei; have a true nucleus; contain many membrane-bound organelles; reproduced by Mitosis and Meiosis.

Examples: Algae, Protozoa, Plants, Animals and Humans.

Other Key Facts

- In both **prokaryotic and eukaryotic cells**, a **semi-fluid matrix** called the **cytoplasm** occupies the cell's volume. The **cytoplasm is a major site of cellular activities** in both plant and animal cells.
- **Ribosomes** are **non-membrane-bound** and present in **both prokaryotic and eukaryotic cells**.
- In prokaryotes, the **nucleus is absent**; genetic material is **naked** and not enclosed by a nuclear membrane. The genomic DNA is typically a single circular chromosome. In addition, many bacteria possess small circular DNA molecules called plasmids, located outside the genomic DNA.



Summary of Prokaryotic & Eukaryotic cell

5. Plant Cell And Animal Cell

Plant cells are the basic unit of life in organisms of the kingdom Plantae. All animal cells are eukaryotic and come from multicellular organisms.

Plant and animal cells exhibit significant structural similarities, as both are classified as **eukaryotic cells**.

Animal Cell	Plant Cell
Are generally small and often irregular in shape.	Generally larger in size than animal cells and have a regular shape.
Do not possess a cell wall and only possess the cell membrane.	Possess cell walls and cell membranes.
Do not possess chloroplasts.	Chloroplasts are present.
Glycogen is stored food material.	Starch is stored food material.
Have centrosome.	Centrosome is present only in some lower plants.
Generally vacuoles are absent and if present, are usually small.	Vacuoles are usually large.

BCW Fact: It is to be noted that the **cell wall is an additional Protective layer outside** the cell membrane of a **plant cell**.

6. Cell Organelles

An organelle is a **subcellular structure** that performs one or more specific jobs in the cell, much like an organ in the body. Each cell contains specific components within it, known as cell organelles. Each kind of cell organelle **performs a special function**.

- The three features or components found in almost every eukaryotic cell are the **plasma membrane, nucleus, and cytoplasm**.

A. Cytoplasm

The Cytoplasm is a gelatinous liquid that fills the interior of the cell. It is **composed of water, salts, and various organic molecules**. It is present within the membrane of cells and embeds all parts of cells and organelles.

- The cytoplasm is the **site of many activities of the cell**.
- The cytoplasm also **assists in metabolic activities**.

B. Cell Envelope

Most prokaryotic cells, particularly bacterial cells, have a chemically complex cell envelope. The cell **envelope often consists of tightly bound three-layered structures**, i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane. These layers work together as a **single protective unit**.

Bacteria are classified into **Gram-positive** and **Gram-negative** based on cell envelope differences and their response to Gram staining. Gram-positive bacteria retain the stain (e.g., Actinomycetes, Bacillus), while Gram-negative bacteria do not (e.g., E. coli, Salmonella).

C. Cell Membrane (Plasma Membrane)

- The plasma membrane is present in all cells (both plant and animal). It is the **outermost covering of the animal cell** that separates the contents of the cell from its external environment.
- Flexible, and is made up of **organic molecules called lipids (mainly phospholipids) and proteins**.
- **Selectively permeable**, it allows or permits the entry and exit of some materials in and out of the cell. Also facilitates processes like **Diffusion and Osmosis**.
- **NOTE:** The cell membrane is absent in the virus.
- **Functions:**
 - Movement of water due to concentration gradient.
 - Active transport of some molecules (energy dependent).

D. Cell Wall

The cell wall is a non-living, rigid outer layer present in plants, fungi, and algae. The Cell wall not only gives **shape to the cell and protects** it from **mechanical damage and infection**, but also helps in **cell-to-cell interaction** and provides a **barrier to undesirable macromolecules**. Algae also have a cell wall.

In plant cells, a cell wall composed of **cellulose** is located outside the cell membrane. Cellulose contributes to the structural strength of plants.

BCW Fact: It is to be noted that the striking difference between a plant and an animal cell is that **“an animal cell lacks a cell wall”**.

E. Mitochondria

Mitochondria are referred to as the **cell's powerhouse**. They generate the energy necessary for various chemical processes vital for life by producing **ATP (Adenosine triphosphate)** molecules. They have two membrane coverings:

- The outer membrane is porous; the inner membrane forms folds called **cristae**.
- Cristae increase surface area for ATP production.
- **ATP** is the cell's energy currency for metabolic and mechanical work.
- ATP (a nucleotide) = Adenine + ribose + three phosphate groups
- Cellular respiration yields up to **38 ATP per glucose molecule** (2 from glycolysis in the **cytoplasm**, 36 from **mitochondrial processes**)
- Mitochondria have their own **DNA** and **ribosomes** → **semi-autonomous** organelles.

BCW Fact: Glycolysis is a sequence of chemical reactions that derive energy from glucose by converting it into two molecules of **pyruvic acid**. This process takes place in the cytoplasm of all living organisms, regardless of whether they are aerobic or anaerobic.

F. Nucleus

The nucleus was **discovered by Robert Brown in 1831**. It has a double-layered covering called the **nuclear membrane**. The nucleus is essential for **cellular reproduction**. In contrast to prokaryotes, eukaryotes have a well-defined nucleus.

- In bacteria, the nuclear region, which lacks a membrane but contains only nucleic acid → called **nucleoid**.
- In non-dividing cells, DNA exists as **chromatin** (thread-like structures).
- During cell division, chromatin condenses to form **chromosomes**.
- Chromosomes carry hereditary information in the form of **DNA** (Deoxyribonucleic Acid) molecules.
- Functional units of DNA are **genes**. DNA stores instructions for cell structure.

1. DNA:

- **DNA** is the hereditary material in humans and almost all organisms.
- Proven genetic material by Alfred Hershey and Martha Chase experiment (1952).
- First identified by Friedrich Miescher in pus cells (1869), and named it '**Nuclein**'.
- **James Watson and Francis Crick** (1953) discovered the **double helical structure of DNA**.
- Made of **nucleotides**: Each composed of sugar groups + phosphate groups + nitrogenous base.

- Four nitrogenous bases of DNA: **Adenine (A) and Guanine (G)** (purines); **Thymine (T) and Cytosine (C)** (pyrimidines).
- Base pairing: **A with T, C with G** → double helix structure.
- Stores information for cell structure and function

2. RNA:

- **RNA** (Ribonucleic Acid) is present in all living cells and helps in **protein synthesis in our body**.
- The four nitrogenous bases of RNA: **Adenine (A), Guanine (G), Cytosine (C), and Uracil (U)**.
- Genetic information flows: **DNA → RNA → Protein** (*Central dogma theory proposed by Crick*).
- Types: **Messenger Ribonucleic Acid (mRNA), Ribosomal Ribonucleic Acid (rRNA), and Transfer Ribonucleic Acid (tRNA)**
- Mostly acts as a messenger in living beings, but as genetic material in some viruses (e.g., Tobacco Mosaic Virus)
- Most enzymes are proteins; some RNAs act as enzymes → **Ribozymes**.

BCW Fact: DNA is present in the nucleus and mitochondria, while RNA is present in the nucleus and cytoplasm.

3. Chromosomes:

Chromosomes are composed of **proteins and DNA (deoxyribonucleic acid)**.

- Chromosomes are made of **DNA and proteins**.
- Humans have **23 pairs (46 total)**
- **22 pairs** are autosomes (same in both sexes). The **23rd pair, the sex chromosomes**, differ between males and females.
- **Female: XX, Male: XY**
- **Klinefelter syndrome:** Male chromosomal disorder with **47 (XXY)** karyotype; caused by an extra X chromosome; may affect physical, behavioral, and cognitive development
- In mammals: **XX = female, XY = male**; In birds/butterflies and moths: **ZW = female, ZZ = male**
- **Gynandromorphism** in birds is rare, caused by chromosomal errors, producing both male and female traits.

G. Endoplasmic Reticulum

Endoplasmic Reticulum (ER) is a network of membrane-bound tubules and sacs.

- Also functions as a cytoplasmic framework, providing a surface for some of the biochemical activities of the cell.
- Two types: **Rough ER (RER)** and **Smooth ER (SER)**.
- **RER** has ribosomes → site of **protein synthesis** → once proteins are produced, they are dispatched to different areas within the cell; Ribosomes are formed in the **nucleolus**.
- **SER** lacks ribosomes and is responsible for the secretion of lipids, which are used for the synthesis of cell membranes. SER is also an essential site for **cholesterol synthesis**.

H. Ribosomes

Ribosomes are the site of protein synthesis. Several ribosomes may attach to a **single mRNA** and form a **chain called polyribosomes or polysomes (protein)**. The ribosomes of a polysome translate the mRNA into proteins, which are the **building blocks of the body**.

- They may be found either in a **free state in the cytoplasm** or **attached to the rough endoplasmic reticulum**.
- In ribosomes, **70S** (prokaryotic) and **80S** (eukaryotic) refer to Svedberg units (S), a measure of **sedimentation rate**.

I. Golgi Bodies

The Golgi apparatus, first described by Camillo Golgi, consists of **membrane-bound vesicles**, which are flattened sacs arranged in parallel stacks called cisternae.

- Golgi membranes are often connected with the **endoplasmic reticulum (ER)**.
- Materials synthesized near the ER are packaged and transported to different destinations inside and outside the cell via the Golgi bodies.
- **Main functions:** storage, modification, and packaging of cellular products into vesicles.
- It can convert simple sugars into complex sugars.
- Plays an important role in the formation of lysosomes.

J. Lysosomes

Lysosomes are small, spherical, membrane-bound vesicles in the cytoplasm ($\approx 0.2\text{--}0.5\ \mu\text{m}$ diameter).

- Contain digestive enzymes and are formed from vesicles budding off the Golgi apparatus.
- Their enzymes are synthesized in the rough endoplasmic reticulum (RER).
- Function as the cell's **"waste disposal system."**
- During metabolic disturbances, lysosomes may rupture, releasing enzymes that digest the cell itself. Hence, called **"suicide bags" or the "atom bomb" of the cell.**
- Common in animal and protist cells; less prominent in plant cells.
- In plants, similar functions are mainly performed by **vacuoles**.

K. Vacuoles

Vacuoles are membrane-bound organelles located within the cytoplasm of the majority of plant, fungal cells, as well as certain animal cells.

- Vacuoles act as **storage sacs** for **solid and liquid** materials.
- In animal cells: small or absent; in plant cells: large and prominent.
- Plant vacuoles contain cell sap, maintaining turgidity and rigidity.
- Store essential substances vital to the plant cell's life, like amino acids, sugars, organic acids, and proteins.
- In some unicellular organisms, specialized vacuoles help remove excess water and waste.

L. Plastids

Plastids are present only in plant cells. They are present in the **cytoplasm** of plant cells. Plastids can be classified into three types:

- **Chloroplasts:**
 - Green plastids containing chlorophyll; site of photosynthesis (mainly in mesophyll cells).
 - Chlorophyll pigments include **chlorophyll a**, **chlorophyll b**, and the yellow pigments (**carotenoids** and **xanthophylls**). Chlorophyll a is the **primary pigment** directly involved in photosynthesis.
 - Pigments located in **thylakoids**.
 - Contain **double-stranded DNA molecules and ribosomes**.
- **Chromoplasts:**
 - Contain fat-soluble carotenoid pigments (**carotene**, **xanthophylls**) Impart yellow, orange, or red colours to plant parts.
 - Help in pollination and seed dispersal.
- **Leucoplasts:**
 - Colourless plastids found in plant cells.
 - Used for the storage of food materials.
 - Amyloplasts – store carbohydrates, mainly starch (e.g., potatoes).
 - Elaioplasts – store oils and fats.
 - Aleuroplasts – store proteins.

M. Peroxisomes

Peroxisomes are **membrane-bound** organelles found in both plant and animal cells, and are especially prevalent in the **green leaves of higher plants**.

- Contain **enzymes for metabolic reactions**
- **Origin:** derived from ER; replicates by **fission**
- **No own genome** (unlike mitochondria/chloroplast)

Functions:

a. In Animal Cells:

- **Oxidation reactions** (via oxidases $\rightarrow$ peroxide formation)
- **Detoxification** (catalase neutralizes H_2O_2 ; alcohol detox in liver)
- **Lipid biosynthesis**

b. In Plant Cells:

- **Fatty acids $\rightarrow$ carbohydrates** (in seeds)

- **Photorespiration** (in leaves)

7. Cell Division

- New cells are formed for growth, repair, replacement, and gamete formation.
- **Cell division** is the process of forming new cells; types: **mitosis** and **meiosis**.

Mitosis:

- A type of **cell division responsible for growth and tissue repair** in organisms.
- One mother cell divides to form **two genetically identical daughter cells**.
- Daughter cells have the **same chromosome number** as the parent cell.
- Occurs in **four stages**: Prophase, Metaphase, Anaphase, and Telophase.

Meiosis:

- Occurs in reproductive cells of plants and animals to form gametes.
- After fertilization, gametes develop into offspring.
- Involves **two successive divisions**: Meiosis I and Meiosis II.
 - **Meiosis I (Reductional division)**: Homologous chromosomes separate; two haploid daughter cells, each containing half the original number of chromosomes.
 - **Meiosis II (Equational division)**: The production of **four new (daughter) cells** rather than the typical two.

Mitosis and Meiosis Differences

Mitosis	Meiosis
• Occurs in all organisms (except viruses)	• Occurs in all sexually reproducing organisms (plants, animals, fungi)
• Produces somatic (body) cells (e.g., blood cells)	• Produces germ/sex cells (e.g., sperm cells)
• Involves one cell division	• Involves two successive divisions
• Produces two diploid (2n) daughter cells	• Produces four haploid (n) daughter cells
• Daughter cells are genetically identical	• Daughter cells are genetically different
• Create a human cell with 46 chromosomes	• Create a human cell with 23 chromosomes

Mitosis and Meiosis Similarity

- Occur in both plants and animals.
- Begin with a **diploid parent cell**.
- Involve **DNA replication**.
- Produce new cells.

BCW Fact:

- In 1839, **Purkinje** introduced the term '**protoplasm**' to describe the **fluid substance within a cell**. **Represents** living cell material, including the nucleus and cytoplasm; mainly composed of **water**, along with proteins, lipids, carbohydrates, and nucleic acids.
- **Ribosomes in Eukaryotes** have a **sedimentation coefficient** of about **80s** and are composed of 2 structural subunits named as **40s** and **60s**, whereas **Prokaryotes** have **70s** and consist of two subunits **30s** and **50s**.
- **Embryology** is the scientific study of embryos and their development. **Karl Ernst von Baer** → "**Father of Modern Embryology**."
- **Ramdeo Misra** → "**Father of Ecology in India**."

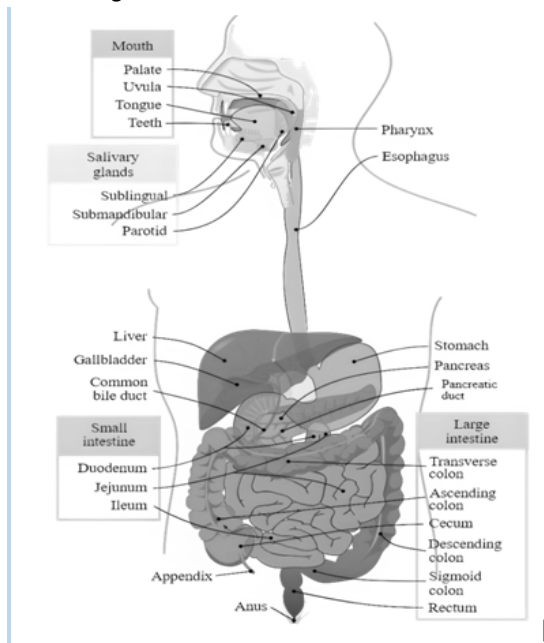
2 - DIGESTIVE SYSTEM

break food into simpler molecules, while absorption transports them for use in the body. Efficient functioning is vital for health, and any disruption may cause malnutrition or digestive disorders.

The human digestive system consists of the **alimentary canal** and the **associated glands**.

1. Alimentary Canal And Associated Glands

Food passes through the digestive system via the **alimentary canal**, a long tube extending from the **mouth to the anus**, with specialized sections performing specific functions. It includes the **mouth, pharynx, oesophagus, stomach, small intestine, large intestine, rectum, and anal canal**. Accessory organs such as the **tongue, teeth, salivary glands, liver, gallbladder, and pancreas** assist in digestion.

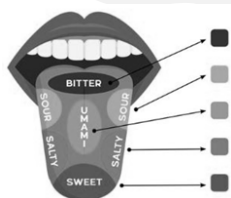


Mouth (Oral/Buccal Cavity)

- The mouth opens into the **buccal or oral cavity**.
- Contains **32 teeth** in adults and a **muscular tongue**.
- Teeth are **thecodont**; enamel is the hardest substance.
- **Types:** I- Incisor-8, C- Canine-4, P- Premolar-8, M- Molar -12.
- The **human dental formula** shows the number and arrangement of teeth in **one-half of each jaw (a quadrant)**. Adult humans have 32 teeth with a formula of 2-1-2-3 (2 incisors, 1 canine, 2 premolars, 3 molars) on each side.
- Humans show **diphyodont dentition** (Temporary (milk) teeth → permanent (adult) teeth).

A. Tongue

- The **tongue** is a highly mobile muscular organ attached to the floor of the oral cavity by the **frenulum**. Its surface bears **papillae**, some containing **taste buds**.



- During chewing, teeth break food into small particles, while the tongue, with saliva, helps in **mastication and mixing** of food.

Taste Bud Regions:

- **Sweet** → present at the tip of the tongue.
- **Salty** → present on either side of the front tongue portion.
- **Sour** → located on the side edges of the tongue.

- **Umami** → located in the middle portion (savoury taste).
- **Bitter** → located at the back (posterior) part of the tongue.

BCW Fact: **Glycogen** is the storage form of carbohydrates in animals, often called **animal starch**. It is mainly found in the **liver, muscles, and brain**. When glucose is required by the body, enzymes convert glycogen into **glucose**. Glycogen is also present in **yeast and fungi**.

C. Salivary Glands

Saliva is secreted by three pairs of salivary glands: the **parotid glands** (cheeks), **submandibular glands** (lower jaw), and **sublingual glands** (beneath the tongue), which release it into the oral cavity.

- **Mucus** lubricates and binds the masticated **food** and helps form a **bolus**.
- Saliva in the oral cavity contains **electrolytes** (Na^+ , K^+ , Cl^- , HCO_3^-) and enzymes like **salivary amylase (ptyalin)** and **lysozyme**.
- **Salivary amylase** begins digestion by breaking **starch into maltose**.
- **Lysozyme** acts as an antibacterial agent, protecting against infection.

D. Pharynx

The oral cavity leads into a short pharynx, which serves as a common passage for food and air.

E. Oesophagus

The oesophagus and the trachea (windpipe) converge at the pharynx. A cartilaginous structure known as the **epiglottis** prevents food from entering the **glottis – the opening of the trachea** – during the act of swallowing.

F. Entry of Bolus to Stomach via Oesophagus

- The **oesophagus** is a thin, long tube that extends through the **neck, thorax, and diaphragm**, and opens into a **J-shaped stomach**. The entry of food from the oesophagus into the stomach is regulated by a **gastro-oesophageal sphincter**.
- Food (bolus) moves through the oesophagus by **peristalsis**, a wave-like muscular contraction.

G. Stomach

The stomach is a muscular, expandable organ that temporarily stores food for about **4–5 hours**.

It receives food from the oesophagus and mixes it with **gastric juice** through churning movements, forming a semi-fluid mass called **chyme**.

Gastric glands in the stomach wall secrete **hydrochloric acid (HCl)**, **pepsinogen (converted to pepsin)**, **mucus**, and **lipase**. In infants, **rennin, a proteolytic enzyme**, helps in the digestion of milk protein (casein).

- **HCl** creates an acidic medium ($\text{pH} \sim 1.8$), activates pepsin, and kills microbes. The hormone **gastrin** regulates HCl secretion and gastric movement.
- **Mucus + bicarbonate** protect the stomach lining and provide lubrication.
- **Pepsin** (from pepsinogen) breaks proteins into peptides (proteoses and peptones), so protein digestion begins in the stomach.
- **Gastric lipase** digests fats into glycerol and fatty acids (limited action).
- Food leaves the stomach gradually through a **pyloric sphincter**, which controls release into the small intestine (Duodenum).

H. Small Intestine

- It is the **longest part of the alimentary canal (about 7.5 m)** and highly coiled.
- Its muscular wall produces movements that **mix food with digestive juices**, aiding digestion.
- **Most digestion** occurs in the **small intestine**.
- It is the site of **complete digestion of carbohydrates, proteins, and fats**.
- Receives secretions from the **liver (bile)** and **pancreas (pancreatic juice)**; also secretes **intestinal juice**.
- These juices act in an **alkaline medium**.
- It is divided into three parts:

- **Duodenum (U-shaped)** – the **shortest part of the** small intestine and the main site of digestion. Here the medium is alkaline.
- **Jejunum** – middle coiled part
- **Ileum** – highly coiled last part; the longest part of the small intestine.
- The length of the small intestines varies in animals:
 - **Herbivores** → require longer intestine (cellulose digestion is slow)
 - **Carnivores** → shorter intestine (meat digests easily)

1. Liver

- Largest gland of the human body; also a major **metabolic center**.
- Produces **bile**, stored in the gallbladder and released into the **duodenum** via the bile duct.
- Bile released into the **duodenum** contains **bile pigments (bilirubin, biliverdin)**, **bile salts**, **cholesterol**, and **phospholipids**, but **no enzymes**. Helps convert acidic chyme into an **alkaline medium** for enzyme action (mainly aided by pancreatic juice).
- Liver is the **largest internal organ**, weighing 1.2–1.5 kg (skin is the largest overall organ).
- Stores **fat-soluble vitamins and glycogen**; acts as a major **detoxification site**; and is the **only organ with significant regenerative ability**.

2. Bile

- Aids digestion of fats by **emulsification** (breaks large fat globules into tiny micelles).
- Also **activates lipase** enzymes.
- Yellowish-green due to **bilirubin and biliverdin**.
- Pigments are formed in the liver from **haemoglobin in dead and worn-out red blood cells (RBCs)**.

3. Gall Bladder

- The **gallbladder** is a small, pear-shaped organ located beneath the liver.
- **Stores and concentrates bile** before releasing it into the small intestine.
- It **receives bile produced by the liver through the common hepatic duct** and is released into the duodenum through the **common bile duct**, aiding fat digestion.
- **Gallstones** form due to insoluble substances like **cholesterol or bilirubin** (from haemoglobin breakdown). Gallstones mainly **disrupt fat digestion**.

4. Pancreas

- Large, cream-coloured gland located **below the stomach**; the **second largest gland** in the body.
- A **mixed gland** with:
 - **Endocrine function**: secretes **insulin and glucagon**
 - **Exocrine function**: secretes **digestive enzymes (pancreatic juice)**
- Produces **pancreatic juice** that works in an **alkaline medium** and helps the digestion of proteins, fats, carbohydrates, and nucleic acids.
- The **small intestine walls** contain glands that secrete **intestinal juice**. Its enzymes complete digestion by converting:
 - **Proteins** → **amino acids**
 - **Carbohydrates** → **glucose**
 - **Fats** → **fatty acids + glycerol**

Regulation of Blood Glucose (Insulin & Glucagon)

- **Carbohydrates** → glucose during digestion → enters bloodstream → **blood sugar rises**, which signals the pancreas to produce insulin.

Insulin (from pancreas):

- Released when **blood glucose increases**.
- Promotes **uptake of glucose by cells** → **lowers blood glucose level**.
- Converts excess glucose into **glycogen** (in liver & muscles).

Glucagon (from pancreas):

- Released when **blood glucose decreases (4–6 hours after meals)**.
- Converts **stored glycogen** (in liver & muscles) → **glucose**.
- Releases glucose into the blood. **Raises** blood glucose level.

- **Pancreatic Juice:** Pancreatic juice comprises six primary types of enzymes that function in an **alkaline environment**:
 - **Amylase** → transforms starch (a polysaccharide) into maltose (a disaccharide);
 - **Lipase (steapsin)** → emulsified fats into fatty acids + glycerol;
 - **Nucleases** → break down nucleic acids such as DNA and RNA found in food;
 - **Trypsinogen** is an inactive form (proenzyme) of trypsin, activated by **enterokinase (enteropeptidase)** from the duodenal lining → active **trypsin** breaks down undigested proteins, proteoses, and peptones into **small peptides**, which are further converted into **amino acids** by other enzymes.
 - **Chymotrypsin** → acts on proteins (including casein) and converts them into **peptides**.
 - **Carboxypeptidase** → breaks peptides into **smaller peptides and amino acids**.

Important Digestive Enzymes

Enzyme	Secreted by	Secreted into	Digestive Juice	Mechanism of action
Salivary amylase (ptyalin)	Salivary glands	Mouth (Oral cavity)	Saliva	• Breaks down starch into maltose.
Pepsin	Gastric glands	Stomach	Gastric juice	• Transforms proteins into peptones and proteoses.
Bile (No enzymes)	Liver	Duodenum	Bile juice	• Emulsification of fats, breaking them down into tiny micelles.
Trypsin	Pancreas	Duodenum	Pancreatic juice	• Breaks down undigested proteins (not processed by pepsin), proteoses, and peptones into smaller peptides, which are subsequently hydrolysed into amino acids by other enzymes.
Amylase	Pancreas	Duodenum	Pancreatic juice	• Transforms starch (a polysaccharide) into maltose.
Pancreatic Lipase	Pancreas	Duodenum	Pancreatic juice	• Transforms fats into fatty acids and Glycerols.
Maltase	Intestinal glands	Small intestine	Intestinal juice	• Breaks down maltose (a disaccharide) into two glucose molecules.
Sucrase	Intestinal glands	Small intestine	Intestinal juice	• Transforms sucrose into glucose and fructose.
Lactase	Intestinal glands	Small intestine	Intestinal juice	• Transforms lactose into glucose and galactose.

NOTE: Small amounts of **lipases** are also secreted by **gastric glands** and **Intestinal glands**.

Absorption (Small Intestine)

- The **intestinal walls absorb digested food**. The inner lining of the small intestine has **villi (finger-like projections)** that increase **surface area for absorption**.
- Villi are richly supplied with **blood vessels**, which transport absorbed nutrients to all body cells.
- In the **duodenum**, food is broken down into simpler forms.
- Absorption mainly occurs in the **jejunum and ileum**.

BCW Fact: The **villi** in the small intestine are provided with a network of capillaries along with a large lymphatic vessel known as the **lacteal**.

I. Large Intestine

- **Undigested/unabsorbed food** passes into the **large intestine (approximately 1.5 m long; shorter than the small intestine)**.
- **No digestion occurs here.**

Functions:

- Absorption of **water, minerals, and some drugs**.
- Secretion of **mucus** for binding and lubrication of waste.
- Feces enter the **cecum** via the **ileo-cecal valve** (prevents backflow). Stored in the **rectum** before **defecation**.
- **Appendix:** vestigial tube attached to the **cecum**.

NOTE: Digestive movements are controlled by **neural mechanisms (both locally and via the Central Nervous System)**. Digestive secretions are regulated by **hormones** from the gastric and intestinal mucosa.

BCW Fact:

- The term '**enzyme**' was coined by **Friedrich Wilhelm Kühne** in 1877 to refer to **biological catalysts**, which were formerly called ferments.
- In the alcoholic fermentation process, **ethyl alcohol** and carbon dioxide are produced from glucose, catalysed by the **zymase** enzyme complex found in yeast.
- **Secretin** is a hormone produced by the S-cells located in the **duodenum** in **response to acidic chyme (containing HCl)** entering from the stomach. Secretin **stimulates the exocrine pancreas**, promoting the release of water and bicarbonate ions.
- **Enterogastrone** → a **hormone produced in the small intestine** in reaction to the **presence of fatty acids**.
- The **functioning of the stomach** → discovered by **William Beaumont**, known as the "**Father of Gastric Physiology**."
- **Ruminants** such as **cattle, buffalo, and deer** rapidly consume and swallow their leafy food, which is **stored in the rumen**. **Cud** refers to the food consumed by ruminants, which is **swallowed and partially digested**. Subsequently, the food is brought back to the mouth where the animal chews it. The **process of nutrition in ruminants** follows a specific sequence: **swallowing → partial digestion → chewing of cud → complete digestion**.

3 - NUTRIENTS AND VITAMINS

1. Nutrients

Nutrients are substances in food required for **body functioning, growth, and maintenance**.

Nutrients are divided into macronutrients (such as **carbohydrates, proteins, fats, and water**) and micronutrients (including vitamins and minerals).

1.1 Carbohydrates

- Macronutrients mainly synthesized by plants; general formula **C_x(H₂O)_y**.
- Examples: **cane sugar, glucose, starch**.

Classification:

- **Monosaccharides:** glucose, fructose, ribose
- **Disaccharides:** sucrose, lactose, maltose
- **Oligosaccharides:** raffinose
- **Polysaccharides:** starch, glycogen, cellulose

Key Points:

- Digested into **glucose**, the main energy source (especially for high-intensity and long-duration activities).
- Excess carbohydrates → converted into **fat** and stored in adipose tissue.
- Carbohydrates can be categorized into **simple and complex**.
 - **Simple carbohydrates** provide quick energy (source-fruits, dairy, vegetables).
 - **Complex carbohydrates:** slow, sustained energy (source- bread, cereals—rice, wheat, bajra, maize, barley, ragi).

Important Examples:

- **Ribose & Deoxyribose:** 5-carbon (pentose) sugars found in RNA and DNA, respectively.
- **Fructose:** sweetest natural sugar; often referred to as fruit sugar (source- fruits, honey).
- **Maltose:** two glucose units (found in cereals).
- **Lactose:** milk sugar.
- Storage forms: **starch (plants)** and **glycogen (animals)**.

1.2 Proteins

- **Proteins** are complex **macromolecules** made of long chains of **amino acids** linked by **peptide bonds**.
- They are essential for the **structure, function, and regulation** of body tissues and organs.

Key Points:

- Composed of **20 amino acids**; some are **essential** amino acids (must be obtained from diet).
- Each amino acid has **—NH₂ (amino)** and **—COOH (carboxyl)** groups with a **variable side chain**.
- Proteins constitute **16–20% of body weight**.
- Most enzymes are proteins.
- **Collagen:** predominant protein in animals.
- **RuBisCO (Ribulose-1, 5-bisphosphate carboxylase/oxygenase):** the most abundant protein in nature and the key enzyme in photosynthesis.

Sources:

- Animal: meat, fish, eggs, milk
- Plant: legumes, soybean, nuts, seeds

Types:

- **Fibrous proteins:** structural (collagen, keratin)
- **Globular proteins:** functional (enzymes, haemoglobin)

Deficiency:

- Kwashiorkor and Marasmus (mainly in children)

1.3 Fats (Lipids)

- **Fats** are a type of **lipid**.
- Composed of **carbon, hydrogen, and oxygen**.
- Compared to carbohydrates, fats have **more carbon & hydrogen** and **less oxygen**.
- Insoluble in water; soluble in organic solvents such as benzene and acetone.
- Dietary fats are mainly **triglycerides**. They are the **most concentrated energy source** and are stored for future energy needs.

Sources:

- Animal: ghee, butter, milk, eggs, meat
- Plant: oils, nuts, seeds

Functions:

- Absorption of **fat-soluble vitamins A, D, E, and K**.

Types of Fats**1. Saturated Fats**

- No double bonds between carbon atoms (saturated with hydrogen)
- Solid at room temperature
- Sources: butter, ghee, coconut oil
- Increase **LDL cholesterol** (risk if excess)

2. Unsaturated Fats

- One or more double bonds in the carbon chain; liquid
- Sources: vegetable oils, nuts, fish
- Lower LDL; heart-friendly

Types:

- **Monounsaturated:** olive oil, avocados, nuts
- **Polyunsaturated:** fatty fish (salmon, mackerel), flaxseeds, walnuts

3. Trans Fats

- **Trans fats** are **unsaturated fats with trans double bonds**, often formed by **partial hydrogenation**.

Sources:

- Industrial: margarine, vanaspati, fried & baked foods
- Natural: meat and dairy from ruminants (cow, sheep, goat)

Health effects:

- Industrial trans fats → Increase LDL and lower HDL (good) cholesterol → higher heart disease risk.
- Natural trans fats → **less harmful in small amounts**.

BCW Fact: The World Health Organization recommended that adults restrict their intake of trans fats to **below 1% of their**

total energy consumption, which is equivalent to less than 2.2 grams per day based on a 2000-calorie diet.

4. **Essential Fatty Acids (EFA)**

- Not synthesized in body.
- Sources: omega-3 (fish oil, flaxseed), omega-6 (soybean oil)
- Important for **brain function, inflammation control, cell function**

Role and Sources of Nutrients

- **Carbohydrates** → energy (rice, wheat, potato, sugarcane)

- **Proteins** → bodybuilding (pulses, eggs, milk, meat)
- **Fats** → energy storage (oils, nuts, ghee)
- **Minerals** → growth (Calcium, Iron, sodium, Potassium, Zinc, etc.)
- **Dietary Fibers (Roughage)** → **elimination of undigested food** (fruits, vegetables, grains)
- **Water** → constitutes approximately **70%** of body weight; essential for **digestion and transport of nutrients, chemicals, and gases** in the body.

2. Vitamins

- Organic compounds required in **small amounts** for growth and metabolism.
- Mostly not synthesized in humans (plants synthesize them).
- Some (e.g., by gut bacteria like *E. coli*) are produced in small amounts.

- Term coined by **Casimir Funk (1912)**.

Functions:

- Act as **catalysts** in energy production by aiding the use of **carbohydrates and fats**.
- Essential for life; **deficiency and excess** both cause health problems.

Types of Vitamins

- **Fat-Soluble Vitamins (A, D, E, K)**: Soluble in fats, not in water; **Stored** in liver and adipose tissue.
- **Water-Soluble Vitamins (B-complex, C)**: Soluble in water; **not stored** in body (except vitamin B12, which is stored in liver); need regular intake as excess is excreted in **urine**.

2.1 Fat-Soluble Vitamins

Vitamin A (Retinol)

- Often called the **anti-infective vitamin** or immunity booster vitamin.
- Acts as **visual pigment + hormone**.
- **β-carotene** (in carrot) = precursor to vitamin A

Functions:

- Essential for **vision (especially in dim light)**.
- Boosts **immunity (anti-infection)**.
- Important for **foetal development**.

Sources:

- Carrot, spinach, kale, mango, papaya, pumpkin, liver, milk, butter, eggs, broccoli, sweet potato

Deficiency:

- **Night blindness or Nyctalopia** (inability to see in dim light)
- **Xerophthalmia** (dry cornea)
- **Keratomalacia** (severe vision loss)

Vitamin D (Calciferol)

- Known as "**sunshine vitamin**" (synthesized in skin via sunlight; **metabolised** in liver & kidney).
- Also called the "**poor man's vitamin**," as it is **freely obtained from sunlight**.

Functions:

- Also called a "**hormone vitamin**" as it **circulates in the blood like hormones** and aids in the **absorption of calcium and phosphorus in the intestine**.
- Essential for **bone health**.
- Supports **immunity and muscle function**.

Sources:

- **Sunlight (main source)**
- Cod liver oil, fish, fish oil

Deficiency:

- **Rickets (children)** → bow legs/knock knees
- **Osteomalacia (adults)** → softening of bones, a disorder of **decreased mineralization of bones**
- **Osteoporosis** → weak and fragile bones (common in women post-menopause)

Vitamin E (Tocopherol or α-Tocopherol)

Functions:

- **Antioxidant** (protects cells from oxidative damage).
- Helps in **RBC formation**.
- Maintains **skin, eyes, and reproductive health**.

Sources:

- Almonds, vegetable oils, wheat germ oil, eggs, whole grains, kiwis, and leafy greens

Deficiency:

- Neuropathy** (reduces the efficiency of nerve impulses)
- Haemolytic anaemia**

Vitamin K

- Derivative of **naphthoquinone**; known as an **anti-hemorrhagic vitamin**.
- Forms: **K₁ (phyloquinone)**, **K₂ (menaquinone)**
- Functions:**
- Essential for **blood clotting** (prothrombin formation with Ca^{2+}).
- Supports **bone metabolism and heart health**.
- Aids **cellular energy production** in mitochondria.
- Synthesized by **intestinal (colon) bacteria**.

Sources:

- Green leafy vegetables (spinach, kale, cabbage), milk, eggs

Deficiency:

- Haemorrhage**, bleeding disorders (bleeding diathesis)

2.2 Water-Soluble Vitamins**Vitamin C (Ascorbic Acid)**

- Very sensitive to heat and temperature.
- Functions:**
- Plays a key role in **collagen synthesis**, essential for **healthy skin, tendons, ligaments, and blood vessels**.
- Enhances **iron absorption**.
- Supports **immunity**.

Sources:

- Citrus fruits, **amla (richest source)**, tomato, spinach, cabbage, papaya, and hot chillies.

Deficiency:

- Scurvy** → bleeding gums, swollen gums, loose teeth, bleeding under the skin.

Vitamin B1 (Thiamine)**Functions:**

- Acts as a **coenzyme in energy metabolism** (carbs, fats, proteins).
- Essential for **nerve function & impulse transmission**.
- Supports **muscle, skin, and hair health**.

Sources:

- Brown rice, whole grains, legumes, nuts, eggs, potatoes, pork, etc.

Deficiency:

- Beriberi** (growth retardation, nerve & muscle issues)

Vitamin B2 (Riboflavin)**Functions:**

- Maintains **healthy skin & blood**.
- Helps in **RBC formation**.
- Essential for **metabolism** (carbs, fats, proteins) & **nervous system**.

Sources:

- Milk, eggs, liver, meat, yeast, green vegetables (beans, asparagus, beet leaves)

Deficiency:

- Cheilosis, glossitis**, mouth inflammation, dermatitis

Vitamin B3 (Niacin)**Functions:**

- Maintains **healthy skin**.
- Important for **energy production & enzyme function**.
- Aids **digestion, appetite regulation, and nerve function**.

Sources:

- Salmon, eggs, milk, nuts, legumes, mushrooms, tomatoes

Deficiency:

- Pellagra** (skin lesions/spots)

Vitamin B5 (Pantothenic Acid)**Functions:**

- Essential for **metabolism** (carbs, fats, proteins).
- Aids **hormone production**.
- Supports **nerve function & impulse transmission**.

Sources:

- Yoghurt, whole grains, meat, legumes, avocado, broccoli, mushrooms, yeast

Deficiency:

- Paraesthesia** (burning sensation)

Vitamin B6 (Pyridoxine)**Functions:**

- Needed for **serotonin formation**.
- Helps in **RBC synthesis**.
- Supports **brain, immune & nerve function**.

Sources:

- Liver, nuts, chickpeas, bananas, cereals, legumes, milk, egg yolk

Deficiency:

- Depression, anaemia, neuropathy**, mouth ulcers, muscular twitching

Vitamin B7 (Biotin)**Functions:**

- Aids **nutrient metabolism** and supports **keratin formation**, helping maintain **healthy skin, nails, and hair**.
- Helps regulate **blood sugar use**.

Sources:

- Spinach, broccoli, egg yolk, liver, nuts, cheese, yeast, meat

Deficiency:

- Dermatitis, hair fall, neuromuscular issues

Vitamin B9 (Folate / Folic Acid)Functions:

- Essential for **DNA & RNA synthesis** (especially in the foetus).
- Supports **nervous system & impulse transmission**.

Sources:

- Green leafy vegetables, legumes, whole grains, liver, sunflower seeds

Deficiency:

- Megaloblastic anaemia

- Spina bifida (in foetus), birth defects

Vitamin B12 (Cyanocobalamin)

- **Water-soluble** but **stored in the liver** (exception).
- Also called **Cyanocobalamin** and contains **cobalt**.

Functions:

- **RBC formation & DNA synthesis**.
- Supports **fat metabolism & central nervous system function**.

Sources:

- Animal products: milk, meat (especially liver), fish, eggs

Deficiency:

- Pernicious anaemia
- neuropsychiatric disorders

BCW Facts:

- **Vitamin K** serves as an **antidote** for anticoagulant poisons, including warfarin and rodenticides, as it reinstates the blood's clotting ability that is inhibited by these poisons.
- **Triglycerides** are **esters** formed from glycerol, a trihydric alcohol, and fatty acids. Typically, triglycerides derived from animal sources are solid, whereas those from plants are predominantly liquid oils. Triglycerides serve as the primary means of **fat storage** in the body. Elevated levels of triglycerides in the bloodstream can heighten the risk of cardiovascular diseases, strokes, and pancreatitis.
- **Gluten** is a family of **proteins** found in wheat, barley, and rye. It is frequently present in products such as bread, pasta, cakes, and beer. **Celiac disease** is an autoimmune disorder that is triggered by the intake of gluten.

4- CLASSIFICATION AND KINGDOM

Earlier characteristics are more primitive than later ones, showing that classification is closely linked to evolution. Most life forms evolved through structural modifications that improved survival. This idea was proposed by **Charles Darwin** in *The Origin of Species* (1859).

In 1758, **Carolus Linnaeus** proposed the **two-kingdom system** (Animalia, Plantae) and introduced **binomial nomenclature**, earning the title "Father of Taxonomy." However, it failed to distinguish prokaryotes/eukaryotes, unicellular/multicellular, and photosynthetic/non-photosynthetic organisms.

In 1969, **R. H. Whittaker** proposed the **Five Kingdom Classification**: Monera, Protista, Fungi, Plantae, Animalia, based on cell structure, nutrition, organization, reproduction, and evolution.

1. Classification

- **Taxonomical classification** groups organisms into categories of ascending order (lowest → highest): **Species → Genus → Family → Order → Class → Phylum/Division → Kingdom**.
- **Species** is the **lowest and most specific category** for all organisms.

Five Kingdom

- **Monera**: Prokaryotic, no nucleus, cellular level
- **Protista**: Unicellular eukaryotes

- **Fungi**: Eukaryotic, chitin cell wall, heterotrophic (saprophytes)
- **Plantae**: Multicellular, cellulose wall, autotrophic
- **Animalia**: Multicellular, no cell wall, heterotrophic

1.1 Monera

- **Only members:** Bacteria; **most abundant microorganisms**.

Cell Structure:

- **Prokaryotic** (lack of a true nucleus or membrane-bound organelles).
- Cell wall is **either present or absent** (e.g., Mycoplasma lacks it).
- **Body Organization:**
- **Unicellular**; no multicellular forms.
- **Mode of Nutrition:**
- Some bacteria → Autotrophic (e.g., cyanobacteria or blue-green algae)
- Most bacteria → Heterotrophic (saprophytic/parasitic; many are disease-causing)
- **Reproduction:**
- Mainly **binary fission**.
- Also shows **primitive genetic exchange**.
- **Examples:**
- Bacteria, cyanobacteria (blue-green algae), mycoplasma
- **Classification Update:**
- Carl Woese divided Monera into:
 - **Archaeobacteria (Archaea)**
 - **Eubacteria (true bacteria)**

BCW Fact: Bacteria are categorized into four types based on their shape: Cocci (spherical or oval cells), Bacilli (rod-shaped cells), Vibrios (comma-shaped), and Spirilla (rigid spiral forms).

1.2 Protista

- **Cell Structure:**
- Mostly **unicellular eukaryotes**; mainly **aquatic**.
- **Body Organization:**
- Movement via **cilia, flagella, or pseudopodia**.
- **Mode of Nutrition:**
- **Autotrophic or heterotrophic**
- **Reproduction:**
- Both **asexual and sexual**; sexual reproduction involves **cell fusion and zygote formation**.
- **Examples:**
- Chlamydomonas, Euglena, Amoeba, Paramecium, Plasmodium, Trypanosoma, Diatoms.

1.3 Fungi

- **Heterotrophic**; commonly found on **moist bread, rotten fruits**.
- Examples: **yeast, moulds, mushrooms, toadstools**

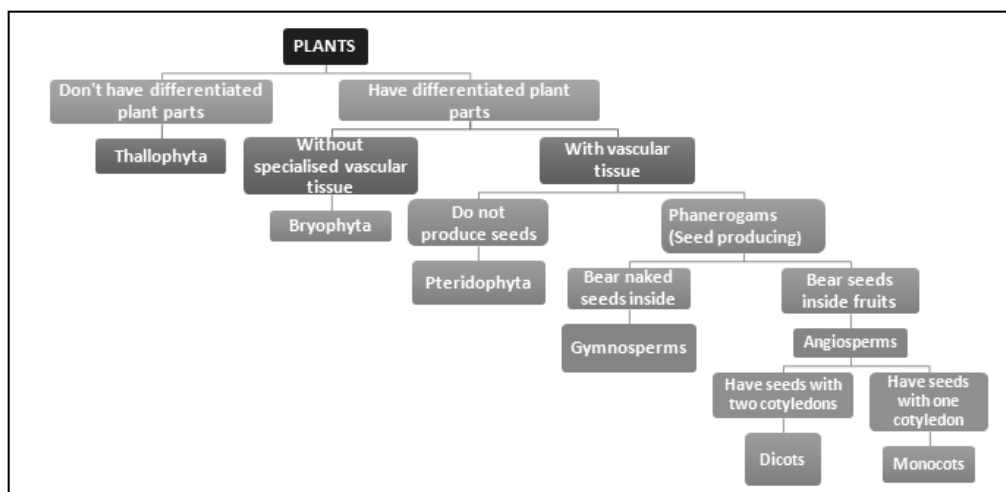
- Most fungi are **filamentous**, made of thread-like **hyphae**; **yeast is unicellular**.
- **Yeast** is used in making **bread, alcohol, wine, and beer**.
- Some fungi are **pathogenic** (e.g., **wheat rust (Puccinia)** and **white spots on mustard leaves**).
- Some are rich sources of **antibiotics**: Penicillium moulds (discovered by Alexander Fleming).
- Fungi are **ubiquitous**—found in **air, water, soil, plants, and animals**.
- **Cell Structure:**
- **Eukaryotic**; uni- or multicellular
- Cell wall made of complex sugar (**chitin**)
- **Body Organization:**
- Many fungi can **shift to multicellular forms** during certain **stages of their life cycle**.
- **Mode of Nutrition:**
- Mostly **heterotrophic saprophytes** (absorb from decaying matter); some **parasites** (on living hosts); also **symbionts**—lichens (with algae) and mycorrhizae (with plant roots) showing **mutualism**.
- **Reproduction:**
- **Vegetative**: fragmentation, budding, fission
- **Asexual**: spores
- **Sexual**: oospores, ascospores, basidiospores

BCW Fact: Ascomycetes are the most common fungi that form a symbiotic relationship with algae to create "lichens". It is to be noted that lichens are very sensitive to atmospheric pollution particularly, sulphur dioxide, and therefore, they are used as **pollution bioindicators**.

1.4 Plantae

Plants are **multicellular eukaryotes** with cell walls. They are autotrophic and perform photosynthesis using chlorophyll. Classification is based on:

- **First level of classification:** Presence of well-differentiated plant body parts.
- **Next level of classification:** Presence of vascular tissues for transport.
- **Further classification:** Ability to produce seeds and whether seeds are enclosed.



1.4.1. Thallophyta

- Plants **lack a well-differentiated body** structure.
- Mostly aquatic** plants and algae.
- Examples:** Spirogyra, Ulva, Chara, Ulothrix, Cladophora.

1.4.2. Bryophyta

- Known as **amphibians of the plant kingdom**.
- Body has stem- and leaf-like parts; attached by rhizoids.
- Lack vascular tissues** (xylem and phloem).
- Examples:** Moss (Funaria), Liverwort (Marchantia).

1.4.3. Pteridophyta

- Body differentiated into **roots, stems, and leaves**.
- First terrestrial plants with **vascular tissues** (xylem and phloem).
- Used as soil-binders and in medicine.
- Examples:** Ferns, Marsilea, Horsetails.

1.4.4. Phanerogams

- Phanerogams** possess well-developed reproductive structures and produce seeds.
- Seeds formed through sexual reproduction contain an embryo and stored food for germination.
- Phanerogams are divided into:
 - Gymnosperms** – naked seeds
 - Angiosperms** – seeds enclosed within fruits

a. Gymnosperms

- "Gymno" = naked, "sperma" = seed.
- Produce naked seeds; mostly **perennial, evergreen** and **woody** plants.
- Examples:** Pine, Cycas, Fir, Spruce, Cedar, Deodar.

b. Angiosperms

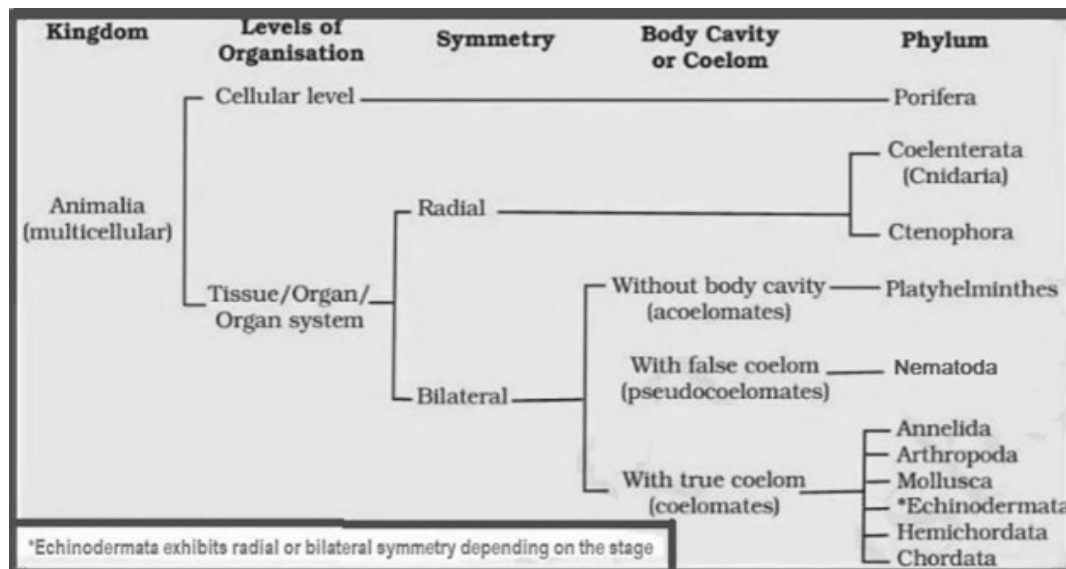
- "Angio" = covered; flowering plants.
- In angiosperms, seeds develop inside the ovary, which later forms the fruit.

- Seed embryos contain **cotyledons (seed leaves)** that may become green during germination.
- Examples:** Mango, Coconut, Pea, Rice, Wheat.
- Based on the cotyledons in the seed, they are classified into two categories:
 - Monocots:** one cotyledon
 - Dicots:** two cotyledons

1.5 Animalia

- Organisms are **eukaryotic, multicellular, and heterotrophic**.
- Cells **lack cell walls**, and most animals are **motile**.
- Classification of animals is based on the following criteria:
 - Level of Organization:** Animals show different levels of organization — **cellular, tissue, organ, and organ-system level**.
 - Body Symmetry:** Body symmetry may be **asymmetrical, radial, or bilateral**.
 - Germ Layers:** Animals may be **diploblastic** (two germ layers) or **triploblastic** (three germ layers).
 - Body Cavity (Coelom):** Classification is based on the presence or absence of a cavity between the body wall and gut wall. A mesoderm-lined body cavity is called a **coelom**. These can be classified as:
 - Acoelomate** – no cavity
 - Pseudocoelomate** – false cavity
 - Coelomate** – true cavity (coelom lined with mesoderm)
 - Segmentation:** Body may be **segmented or unsegmented**.
 - Notochord:** A rod-shaped structure derived from mesoderm; present in **chordates** and absent in **non-chordates**.

Further classification is based on the **degree and nature of body organization**.



2.5.1 Porifera

- “**Porifera**” = **pore-bearing animals** with a canal system for water, food, and oxygen circulation.
- Commonly called **sponges**; mostly **marine and asymmetrical**.
- **Non-motile** and **attached to solid surfaces**.
- Reproduce both **asexually** (fragmentation) and **sexually** (gametes).
- **Examples:** *Sycon (Scypha)*, *Spongilla*, *Euspongia* (bath sponge).

2.5.2 Coelenterata (Cnidaria)

- **Aquatic**, mostly **marine**; may be **sessile or free-swimming**.
- **Radially symmetrical** and **diploblastic** with a body cavity.
- Exist as **colonial** (corals) or **solitary** forms (*Hydra*).
- **Examples:** Jellyfish, Sea anemone, Coral, Sea-fan.

2.5.3 Platyhelminthes

- **Bilaterally symmetrical, triploblastic, and acoelomate**.
- Body is **dorsoventrally** (top to bottom) **flattened** → called **flatworms**.
- Some are free-living, many are **parasitic**.
- **Examples:** *Planaria*, *Liver fluke*, *Tapeworm*.

2.5.4 Nematoda

- **Bilaterally symmetrical and triploblastic**.
- Cylindrical body with **pseudocoelom** (false body cavity).
- Organ level absent; many are **parasitic**.
- **Examples:** *Ascaris*, *Wuchereria*, *Roundworm*, *Pinworm*.

2.5.5 Annelida

- **Bilaterally symmetrical, triploblastic, and coelomate**.
- Body is **segmented** with well-developed organs.
- Found in **marine, freshwater, and terrestrial habitats**.
- **Examples:** Earthworm, Nereis, Leech.

2.5.6 Arthropoda

- **Largest animal phylum**.
- **Segmented body**, bilateral symmetry, and **jointed legs**.
- **Open circulatory system**; Coelomic cavity filled with blood.
- **Examples:** Cockroach, Butterfly, Spider, Crab, Scorpion.

2.5.7 Mollusca

- **Bilaterally symmetrical with reduced coelom**.
- Body soft, with **muscular foot** for movement.
- **Open circulatory system** and kidney-like excretory organs.
- **Examples:** Snail, Octopus, Mussel, Squid, Cuttlefish.

2.5.8 Echinodermata

- **Marine, spiny-skinned animals**.
- **Triploblastic and coelomate with water vascular system**.
- Skeleton made of hard **calcium carbonate**.
- **Examples:** Starfish, Sea urchin, Sea cucumber.

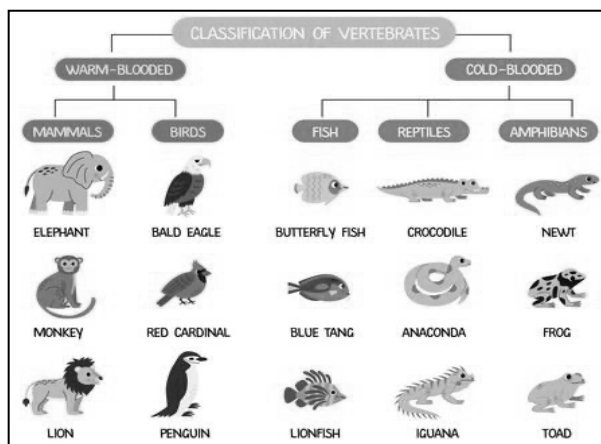
2.5.9 Hemichordata

- **Marine, worm-like, coelomate animals**.
- Bilateral symmetry with **organ-system level** organization.
- Respiration through **gills**; excretion by **proboscis gland**.
- **Examples:** *Balanoglossus*, *Saccoglossus*.

2.5.10 Chordata

- **Notochord** is present at some stage of life.
- Possess **dorsal hollow nerve cord, pharyngeal gill slits, and post-anal tail.**

- **Bilaterally symmetrical, triploblastic, coelomate with a closed circulatory system.**
- Divided into **Urochordata, Cephalochordata, and Vertebrata.**



Vertebrata

- Vertebrates are characterized by the presence of a **true vertebral column** and an **internal skeleton**, which facilitates a distinct arrangement of muscle attachment points for locomotion.
- Vertebrates are classified into five categories: **Aves, Pisces, Amphibia, Mammalia, and Reptilia.**

i) Aves

- Body covered with feathers.
- All birds are included in this classification.
- Most can fly (flightless birds like ostrich are exceptions).
- **Warm-blooded (endothermic). Four-chambered heart.**
- **Examples:** Penguin, Parrot, Peacock, Vulture, Ostrich.

ii) Pisces

- Aquatic animals; Skin covered with scales or plates.
- **Breathe through gills** (oxygen from water).
- Cold-blooded (ectothermic); **Two-chambered heart.**
- Mostly reproduce by laying eggs (oviparous).
- **Examples:** Rohu, Sea-horse, Flying Fish, Climbing perch.

iii) Amphibia

- Moist, scale-less skin.
- **Cold-blooded (ectothermic); Heart with three chambers.**
- Respiration through **gills, lungs, and skin.**

- Lay eggs (oviparous).
- **Live both in water and on land** (aquatic + terrestrial).
- **Examples:** Common frogs, Toads, Hyla (Tree frog), and Salamanders.

iv) Mammalia

- Warm-blooded (endothermic); **Four-chambered heart.**
- Possess **mammary glands (produce milk for young).**
- Skin covered with hair;
- Most give birth to **live young (viviparous).**
- Some lay eggs (e.g., platypus, echidna).
- Some give **birth to underdeveloped young (e.g., kangaroo).**
- **Examples:** Blue whale, Human, Elephant, Cat, Dog, Camel, Common dolphin.

v) Reptilia

- Cold-blooded (ectothermic).
- Creeping or crawling mode of locomotion.
- Respiration through lungs. Mostly **terrestrial.**
- **Heart with three chambers (except crocodiles – four chambers).**
- No external ear openings; **tympanum** acts as ear.
- Usually have two pairs of limbs (if present).
- **Examples:** Tortoise, Turtles, Lizards, Alligator, Chameleon, King Cobra.

BCW Facts:

- An **earthworm** is classified as one of the animals within the **Annelida phylum**. In earthworms, the **blood vascular system is of a closed type**. The blood of an earthworm is red, and **haemoglobin is dissolved in the blood plasma**. They do not

possess **red blood cells**.

- **Method of reproduction** - Planaria (Regeneration), Amoeba (Binary fission), Spirogyra (Fragmentation), Yeast (Budding).
- In Whittaker's five-kingdom classification, **lichens** and certain acellular organisms such as **viruses, viroids, and prions** are not included.
- **Lichens** represent symbiotic relationships, meaning mutually beneficial associations, between **algae and fungi**. The algal part is referred to as **phycobiont**, while the fungal part is called **mycobiont**, with the former being autotrophic and the latter heterotrophic.
- **Gambusia fish** are introduced into ponds to feed on **mosquito larvae**, effectively controlling their populations.

5- MICROBES AND DISEASES

1. Microbes And Their Uses

Microbes include **bacteria, viruses, fungi, protozoa, and some algae**. While many are beneficial—helping in fermentation, decomposition, and nitrogen fixation—some cause diseases in humans, animals, and plants. Some microbes can be harmful, leading to diseases in humans, animals, and plants.

- **Microorganisms** such as amoebae can live independently, while many fungi and bacteria commonly grow in colonies.
- Microbes are used in the preparation of **curd, bread, and cakes**. Yeast is also used in the commercial production of **alcoholic beverages, such as beer and wine**.
- Many antibiotics are produced by microorganisms. **Streptomycin, tetracycline, and erythromycin** are well-known antibiotics produced by bacteria and fungi.
- The bacterium **Rhizobium** plays an important role in **nitrogen fixation** in the root nodules of leguminous plants (pulses).

BCW Fact: Cyanobacteria (blue-green algae) are **autotrophic microorganisms** found widely in aquatic and terrestrial ecosystems. Many cyanobacteria can **fix atmospheric nitrogen**. Common examples include **Anabaena, Nostoc, and Oscillatoria**.

1.1. Bacterial Diseases

Bacteria are responsible for causing many infectious diseases like pneumonia, tuberculosis, diphtheria, syphilis, and tooth decay. There are a variety of bacteria that cause diseases in humans:

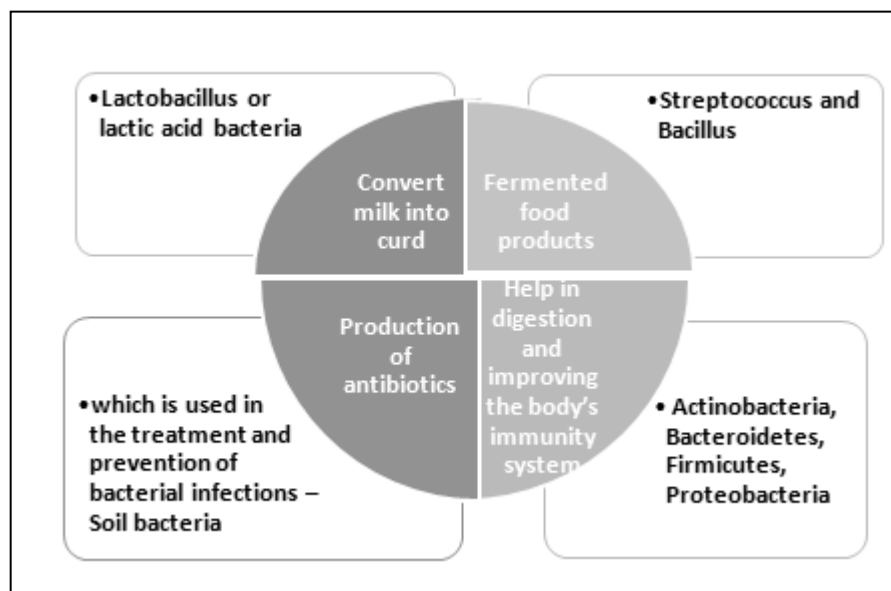
S.No.	Disease Name	Description
1.	Dental Caries (Tooth Decay)	Caused by bacteria producing acids that damage tooth enamel and form plaque ; leads to cavities , pain, infection, and possible tooth loss if untreated.
2.	Typhoid	Caused by bacterium Salmonella Typhi ; spread through contaminated food and water ; Widal test is commonly used for diagnosis.
3.	Food poisoning	Sometimes, microorganisms grow on the food we consume. Food becomes contaminated when these microorganisms multiply and release toxic substances, causing food poisoning. Foodborne botulism is caused by the bacterium Clostridium botulinum.
4.	Cholera	Caused by bacterium Vibrio cholerae ; spread through contaminated food or water . Infected persons can spread bacteria through stool for 7–14 days .
5.	Tetanus	An infection caused by bacteria called Clostridium tetani ; produces toxin causing painful muscle contractions. Commonly affects jaw and neck muscles .
6.	Tuberculosis	Caused by Mycobacterium tuberculosis ; mainly affects the lungs (Pulmonary TB), but may affect the kidney, spine, and brain. Spread through air droplets released during coughing, sneezing, or spitting; people with HIV/AIDS and others with weakened immune systems are more susceptible. TB Diagnosis & Prevention: GeneXpert test and Mantoux test are used for diagnosis; BCG vaccine protects against tuberculosis and also provides some protection against leprosy ;

		First-line anti-TB drugs: Isoniazid + Rifampicin .
7.	Diphtheria	An acute bacterial infection/disease caused by Corynebacterium diphtheriae (C. diphtheriae) , primarily affecting the tonsils, pharynx, and nose. Common symptoms - sore throat, fever, swollen neck glands, and weakness. Prevented by vaccination and booster doses.
8.	Whooping Cough (Pertussis)	Caused by the bacterium Bordetella pertussis ; highly infectious respiratory disease spread through droplets , most dangerous for infants . Prevented by DTP (Diphtheria-Tetanus-Pertussis) vaccine .

NOTE: Typhoid and Cholera are water-borne diseases.

Useful Bacteria

Not all bacteria are harmful to humans. There are some bacteria that are beneficial in different ways.



1.2. Protozoan Diseases

Malaria

- A **mosquito-borne protozoan infectious disease** that directly affects both animals and humans.
- Caused by **Plasmodium** protozoan, and the vector is female **Anopheles mosquitoes**.
- **Plasmodium falciparum** (most lethal species) and **Plasmodium vivax** account for the majority of malaria cases.
- **Symptoms:** fatigue, vomiting, fever, and headache. These two types are commonly reported from India.
- Earlier treated with **Quinine** (made from Cinchona stem bark); now **Chloroquine phosphate** (preventive/therapeutic).
- The most affected organ of body: **Spleen**.

Spleen

- The spleen is a complex, **secondary**, and the **largest lymphoid organ**, which plays a crucial role in controlling **blood-stage infection** with *Plasmodium* parasites.

- The primary function of the spleen is to **filter blood** by **eliminating cellular waste and getting rid of old, malformed, or damaged red blood cells**.
- It aids in the fight **against bacteria** that cause illnesses.

Kala-azar (Visceral Leishmaniasis)

- Caused by the protozoan parasite **Leishmania donovani**.
- Transmitted by **sandflies (Phlebotomus argentipes)**.
- Affects the **Reticuloendothelial System (RES)**, and is also found in the **bone marrow, liver, and spleen**.

Amoebiasis (Amoebic Dysentery)

- Caused by **Entamoeba histolytica** in the **large intestine**.
- Symptoms: **abdominal pain, cramps, constipation, and mucus- & blood-containing stools**.
- Spread through contaminated food and water; **houseflies act as mechanical vectors**.

Sleeping Sickness (Human African Trypanosomiasis)

- Caused by protozoa of the genus **Trypanosoma**.

- Transmitted by the bite of infected **tsetse flies (Glossina)**.
- Common in **sub-Saharan Africa**.

1.3. FUNGAL DISEASES

Some common fungal diseases are athlete's foot, dermatitis, ringworms, etc.

Athlete's Foot (Tinea Pedis)

- A **fungal skin infection** that usually develops **between the toes**.
- Common in people with **sweaty feet** wearing **tight-fitting shoes**.

Candidiasis

- A fungal infection caused by **Candida yeasts**.
- Commonly affects the **mouth, vagina, skin, oesophagus, and bloodstream**.
- **Vulvovaginal candidiasis** is one of the most common forms.

Ringworm

- A fungal infection caused by **Microsporum, Trichophyton, and Epidermophyton**.
- Produces a **red, itchy, ring-shaped rash**.
- Symptoms include **dry, scaly lesions on the skin, nails, and scalp**.

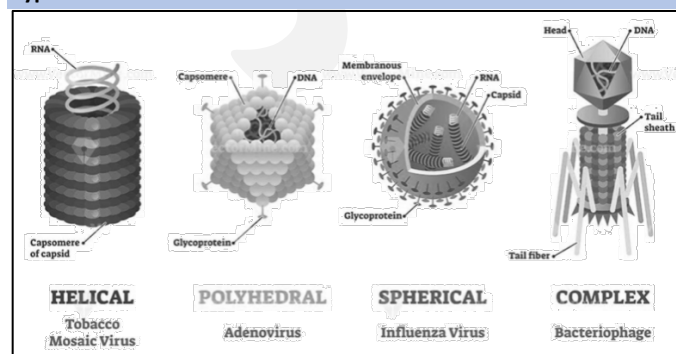
- **Aflatoxins** are toxic substances produced by the fungi **Aspergillus flavus** and **Aspergillus parasiticus**. They commonly contaminate crops such as **maize (corn), peanuts, cottonseed, and tree nuts**, especially in **warm and humid regions**.
- **Mycorrhiza** is a **symbiotic association** between a **fungus** and **plant roots**. The fungal filaments surround or penetrate the roots, increasing the **surface area for absorption**. The roots obtain **water and minerals** from **fungi**, while the **mycorrhizae** obtain **sugars and nitrogen-containing components** from the roots.

BCW Fact: Liver flukes are **parasitic flatworms** that infect the **liver and bile ducts** of humans and other mammals. **Fasciola hepatica**, commonly called the **Sheep Liver Fluke**, is mainly found in the bile ducts of **sheep, goats, cattle, deer, and other grazing animals** in wet or marshy areas. Humans can get infected by **eating raw or undercooked fish, crabs, or crayfish** from areas where the parasite is found, or infected by consuming **raw water plants (such as watercress)** contaminated with parasite larvae.

1.4. Viruses

- Viruses are not included in biological classification because they are **not regarded as genuinely 'living'** due to their absence of a cellular structure.
 - The viruses are **non-cellular organisms** that are characterised by having an **inert crystalline structure outside the living cell**.
 - A virus refers to a **toxic substance or poisonous fluid**.
 - **Dmitri Ivanowsky (1892)** discovered that the agent causing **tobacco mosaic disease** could pass through **bacteria-proof filters**, showing it was smaller than bacteria.
 - **M.W. Beijerinck (1898)** proved the infectious agent could reproduce in healthy plants, named it a **virus**, and called it **Contagium vivum fluidum** ("infectious living fluid").
 - **W.M. Stanley (1935)** demonstrated that viruses can be **crystallized** and are composed mainly of **proteins**.
- They are inert outside their specific host cell. Viruses are obligate parasites.

Types of Viruses



- Viruses contain **genetic material (RNA or DNA)** enclosed in a **protein coat**; no virus possesses both RNA and DNA.
- A virus consists of **nucleoprotein** whose genetic material is **infectious**.
- **Plant viruses** generally possess **single-stranded RNA**, while **animal viruses** may contain **single-stranded RNA**, **double-stranded RNA**, or **double-stranded DNA**.
- **Bacteriophages** are usually **double-stranded DNA viruses** that infect bacteria.
- Viruses cause diseases such as **mumps, smallpox, herpes, influenza, and AIDS**.
- In plants, viral infections cause **mosaic patterns, leaf rolling and curling, yellowing of veins, dwarfism, and stunted growth**.
- Viruses cannot grow in **synthetic environments** (i.e., they cannot be cultured) as they lack **energy-producing systems** and depend entirely on **host cells**.

Diseases Caused By Viruses

Polio (Poliomyelitis)

- **Poliovirus** belongs to the **Enterovirus** genus of the **Picornaviridae** family and spreads mainly through the **faecal-oral route**.
- It mainly affects **children below 5 years** and can invade the **nervous system**, causing **irreversible paralysis**.
- Three strains of wild poliovirus:
 - **Wild Poliovirus type 1 (WPV1)** – active strain
 - **Wild Poliovirus type 2 (WPV2)** – eradicated in 2015
 - **Wild Poliovirus type 3 (WPV3)** – eradicated in 2019
- Symptoms include **fever, weakness, muscle paralysis**, and, in severe cases, **respiratory failure**.
- **No cure exists**, but prevention is possible through **immunisation**.

Vaccines

- **OPV (Oral Polio Vaccine)** – developed by Albert Sabin; administered orally.
- **IPV (Injectable Polio Vaccine)** – given under the **Universal Immunisation Programme (UIP)**.

BCW Fact: World Polio Day is celebrated on 24th October to raise awareness of the importance of polio vaccination

and commemorate the **birth anniversary of Jonas Salk**. He was the first person to lead a team that developed the **Inactivated (killed) Polio Vaccine (IPV)** against the disease.

Hepatitis

- **Hepatitis** is inflammation of the **liver** caused by **viral or non-viral agents**.
- Major hepatotropic viruses: **Hepatitis A, B, C, D, and E**.
- **Hepatitis B and C** commonly cause **chronic liver disease, cirrhosis, and liver cancer**.
- **Hepatitis D** occurs only in the presence of **Hepatitis B virus (HBV)**.

Transmission

- **Hepatitis A & E** – spread through **contaminated food and water**.
- **Hepatitis B, C & D** – spread through **infected blood/body fluids**, unsafe injections, sexual contact, and mother-to-child transmission.

Key Facts

- **Hepatitis B virus** is a **partially double-stranded DNA virus**.
- Vaccines are available for **Hepatitis A and B**.
- **No vaccine** exists for **Hepatitis C** due to multiple viral subtypes.
- Chronic Hepatitis C is treated using **antiviral therapy**.

Viruses Affecting the Brain

a) Meningitis

- An infection and inflammation of the fluid and membranes surrounding the brain and spinal cord. These membranes are called **meninges**.
- **Symptoms:** headache, fever and a stiff neck.

b) Japanese encephalitis

- A **mosquito-borne viral disease** caused by Japanese encephalitis virus (JEV).
- The Japanese encephalitis virus (JEV) is a flavivirus that is associated with dengue, yellow fever, and West Nile viruses, and it is transmitted by mosquitoes, **particularly Culex tritaeniorhynchus**.

Parkinson's disease

Parkinson's disease is a **progressive disorder of nerve cells** in the part of the brain called *the substantia nigra*, which controls movement. The dying/weakening of these nerve cells leads to symptoms such as tremors in the hands or arms, and unsteady walking, et. Named after the English scientist James Parkinson.

Arvid Carlsson was a renowned Swedish neuropharmacologist, primarily recognized for his research on the neurotransmitter

dopamine and its implications for Parkinson's disease. In recognition of his contributions to the understanding of dopamine, Carlsson received the **Nobel Prize in Physiology or Medicine** in 2000, sharing this honour with Eric Kandel and Paul Greengard.

Other Mosquito-Borne Viral Diseases

Dengue

- Spread through the bites of infected female mosquitoes, mainly the *Aedes aegypti* and *Aedes albopictus*.
- It can lead to a significant **drop in blood platelet count**.

Yellow fever

It is **prevalent in parts of Africa and South America**, and is spread primarily by the *Aedes aegypti* mosquito. Symptoms include **nausea, vomiting and jaundice**.

Zika

- A viral disease, spread by the *Aedes* mosquito (mainly *Aedes aegypti*), which bites during the day.
- **Symptoms:** fever, rash, joint pain, and conjunctivitis (red eyes).

Chikungunya

- A viral disease transmitted by *Aedes aegypti* and *Aedes albopictus* mosquitoes, resulting from the chikungunya virus (CHIKV), which is an RNA virus belonging to the alphavirus genus within the **Togaviridae** family.
- **Symptoms:** severe joint pain, rashes, headache, fever, nausea, vomiting, and muscle pain.

Different Species of Mosquitoes as Disease Vectors

- *Anopheles* mosquitoes transmit **malaria** and are mainly found in the **Indian subcontinent, Southeast Asia, and the Asia-Pacific region**.
- *Culex* mosquitoes transmit **Japanese encephalitis** and **West Nile virus**. They are found worldwide except in colder temperate regions and are common in **Los Angeles**.
- Adult *Anopheles* rest at a **45° angle to the surface** they are standing on, whereas *Aedes* and *Culex* rest with their bodies **parallel to the surface**.
- *Anopheles* larvae lie **horizontally** on the water surface, while *Aedes* and *Culex* larvae hang at an **angle below the surface**.

Ebola Virus Disease (EVD)

- **EVD**, formerly called **Ebola haemorrhagic fever**, is a highly deadly disease with outbreaks mainly in **Africa**.
- The **Ebola virus** was first identified in **1976** near the **Ebola River** in the Democratic Republic of the Congo.
- The virus spreads from wild animals such as **fruit bats, porcupines, monkeys, gorillas, and chimpanzees** to humans.

Transmission & Symptoms

- **Animal-to-human transmission:** Through contact with the blood, organs, secretions, or bodily fluids of infected animals.
- **Human-to-human transmission:** Through direct contact with infected body fluids or contaminated objects like clothes and bedsheets.
- The virus enters the body through **skin cuts** or contact with the **eyes, nose, or mouth**.
- Early symptoms include **fever, fatigue, and headache**.
- The incubation period ranges from **2–21 days**.

Diagnosis & Vaccine

- Diagnosed through **ELISA** and **antigen-capture detection tests**.

- The **Ervebo (rVSV-ZEBOV)** vaccine has shown high effectiveness against Ebola.

HIV and AIDS

- **HIV** is a **lentivirus** (subclass of retroviruses) with **RNA** as genetic material. It causes **HIV infection**, which may progress to **AIDS (Acquired Immunodeficiency Syndrome)**.
- India aims to eliminate AIDS by **2030**.
- HIV infection is detected through **ELISA** and **Western Blot** tests.
- Symptoms may appear within **2–4 weeks** of infection, while progression to AIDS generally takes **5–10 years** without treatment.

Mechanism

- HIV attacks **CD4+ T cells (helper T-lymphocytes)**, weakening the immune system.
- It multiplies inside the body and progressively destroys CD4 cells.
- In healthy individuals, CD4 count ranges from **500–1600**; in HIV patients, it may fall below **200**, which is diagnosed as **AIDS**.

- Once HIV enters the body, it cannot be completely removed.
- Transmission**
- Spread through infected **blood, breast milk, semen, vaginal fluid, and pre-ejaculate**.
- Symptoms**
- Swollen lymph nodes, weight loss, fever, diarrhoea, memory loss, and night sweats.**
- Weak immunity increases susceptibility to **opportunistic infections and cancers**.
- Treatment & Prevention**
- HIV can be managed and prevented through **Antiretroviral Therapy (ART)**.
- NOTE:** Sweat, saliva, tears, urine, faecal matter, and mosquito bites do not cause AIDS, as they don't have Blood Stains.

Pneumonia

- Pneumonia can be caused by **viruses, bacteria, or fungi**.
- Pneumonia represents a type of acute respiratory infection that impacts the lungs. The lungs consist of tiny sacs known as **alveoli**, which expand with air when a healthy individual inhales. In cases of pneumonia, the alveoli become filled with pus and fluid, resulting in painful breathing and restricted oxygen absorption.
- Streptococcus pneumoniae** is the leading cause of bacterial pneumonia among children. **Haemophilus influenzae type b (Hib)** ranks as the second most prevalent cause of bacterial pneumonia.
- The **respiratory syncytial virus** is the primary viral agent responsible for pneumonia.
- In infants infected with HIV, the fungus **Pneumocystis jirovecii** is among the most frequent causes of pneumonia.
- Antibiotics should be utilized for the treatment of pneumonia.

Vaccination

- Edward Jenner is regarded as the **father of immunology** for developing the world's first successful **smallpox vaccine** in 1796.
- Jenner observed that **cowpox** infection protected milkmaids from **smallpox**.
- He inoculated **8-year-old James Phipps** with material taken from a cowpox sore on milkmaid **Sarah Nelmes**.
- Phipps developed mild illness but recovered fully. Later exposure to smallpox did not infect him, proving immunity.
- Jenner's work laid the foundation of **modern vaccination and immunology**.
- A **vaccine** is a biological preparation that provides **acquired immunity** against a specific disease.
- It contains weakened or killed microorganisms, toxins, or surface proteins resembling the disease-causing pathogen.
- Vaccines stimulate the immune system to recognize, destroy, and remember the pathogen for future protection.
- Live attenuated vaccines**
- Pathogens such as viruses or bacteria are weakened through genetic modifications to restrict their growth, thereby preventing disease in the host.
- In certain modified versions of live vaccines, an organism closely related to the pathogen is utilized, which naturally has limited growth in humans.
- Examples:** Nasal influenza vaccine, Varicella vaccine, Bacille Calmette-Guerin (BCG) vaccine, Rotavirus vaccine.
- Inactivated or Dead vaccines**
- The pathogen responsible for the disease is either killed or inactivated, typically through thermal (high temperature application) or chemical (such as formalin) methods.
- Examples:** Salk polio vaccine, Typhoid vaccine, Hepatitis A vaccine.
- Subunit, recombinant, polysaccharide, and conjugate vaccines**
- Subunit vaccines do not contain whole cells. Instead, they consist of polysaccharides or proteins derived from the surface of the bacteria or virus.

BCW Fact: Smallpox is caused by the **variola virus**, a member of the **Orthopoxvirus family**, and was officially declared **eradicated** by the World Health Organization in **1980**.

Types Of Vaccines

Vaccines can be synthesized in various ways, which leads to their classification:

There are subtypes of Subunit vaccines:

A. Conjugate vaccines

- Conjugate vaccines connect polysaccharides to proteins to enhance the immune response, particularly in young children.

- **Examples:** Pneumococcal conjugate vaccine, Meningococcal C conjugate vaccine, Haemophilus influenzae type b (Hib) conjugate vaccine.

B. Recombinant Vaccines

- Recombinant vaccines represent a subtype of subunit vaccines that employ genetic engineering to create specific pathogen antigens, thus avoiding the use of the entire microbe.

- **Examples:** Hepatitis B vaccine, Human papillomavirus (HPV) vaccine.

Toxoid Vaccines

- These vaccines utilize inactivated toxins (toxoids) produced by bacteria to stimulate immunity against the toxin rather than the microbe itself.

- **Examples:** Diphtheria vaccine, Tetanus vaccine.

mRNA Vaccines

- These vaccines employ messenger RNA to direct cells to generate a pathogen protein antigen.

- **Examples:** COVID-19 vaccines (Pfizer-BioNTech, Moderna).

Viral Vector Vaccines

- Viral vector-based vaccines are distinct from the majority of traditional vaccines because they do not contain antigens themselves; instead, they utilize the body's own cells to generate them.

- This is achieved by employing a **modified virus (the vector)** to introduce genetic instructions for the antigen, specifically the COVID-19 spike proteins located on the virus's surface, into human cells.

- **Examples:** COVID-19 vaccines (AstraZeneca, Johnson & Johnson) and the Ebola vaccine.

Zoonotic Diseases

Plague

- Plague is an infectious disease caused by the bacterium **Yersinia pestis**, a zoonotic bacterium found in small mammals and their fleas.
- Transmission occurs through **infected flea bites** or handling infected animals.
- The most common form is **bubonic plague**, characterized by swollen and painful lymph nodes called **buboes**.

Brucellosis

- Brucellosis is a zoonotic disease caused by the bacterial genus **Brucella**.

- It mainly infects **cattle, swine, goats, sheep, and dogs**.

- Humans contract it through **direct contact with infected animals**, contaminated animal products, or inhalation of airborne agents.

Anthrax

- Anthrax is caused by the spore-forming bacterium **Bacillus anthracis**.

- It mainly affects ruminants such as **cows, sheep, and goats** and can spread to humans through infected animals or contaminated animal products.

- The bacteria produce potent toxins responsible for severe symptoms and high lethality.

- It is also called **splenic fever** due to enlargement of the spleen.

Rabies

- Rabies is an acute viral disease causing **encephalomyelitis** in warm-blooded animals, including humans.

- In developing countries, **dogs** are the main reservoirs, while in developed countries, wild animals such as **foxes, raccoons, and coyotes** are common reservoirs.

Nipah

- **Nipah virus** is a zoonotic virus transmitted from animals such as **bats and pigs**, contaminated food, or through human-to-human spread.

- It was first identified in **1999** during an outbreak among pig farmers in Malaysia.

Swine Flu

- Swine Flu initially emerged from pigs. It is caused by a virus that impacts pigs and has begun infecting humans.

- The **Influenza virus H1N1** has been recognized as the source of this illness.

Avian Influenza

- Avian influenza is a transmissible disease among birds caused by the **Influenza Virus H5N1**. Birds capable of carrying and disseminating the avian influenza virus include poultry (such as chickens, turkeys, or ducks), wild birds, and pet birds.

Common Diseases

Disease	Type	Caused By
Chickenpox	Viral	Human herpes virus (Varicella zoster)
Influenza	Viral	Influenza viruses
Measles	Viral	Paramyxoviruses
Chikungunya	Viral	Chikungunya Virus
Herpes	Viral	Herpes Simplex Virus (HSV)

Mumps	Viral	Paramyxovirus
Rabies	Viral	Rhabdovirus
HPV	Viral	Human Papillomavirus
Tetanus	Bacterial	Clostridium tetani
Gonorrhoea	Bacterial	Neisseria gonorrhoeae
Tuberculosis	Bacterial	Mycobacterium tuberculosis
Typhoid	Bacterial	Salmonella typhi
Plague	Bacterial	Yersinia pestis
Diphtheria	Bacterial	Corynebacterium diphtheriae
Cholera	Bacterial	Vibrio cholerae

Genetic Disorder

- **Down syndrome** results from the **trisomy of chromosome 21**, which means there is an additional copy of chromosome 21, leading to a total chromosome count of 47.
- In the case of **Turner's syndrome**, one X chromosome is absent, resulting in a karyotype of 45, XO, while in **Klinefelter's syndrome**, there is an extra copy of the X chromosome, resulting in a karyotype of 47, XXY.
- **Thalassemia** is an **autosomal recessive blood disorder** that is inherited from parents to their children when both parents are carriers of the gene (or heterozygous).
- **Colour Blindness** is a **sex-linked recessive disorder** caused by a defect in either the red or green cones of the eye, leading to an inability to distinguish between red and green colours. The defect arises from mutations in specific genes located on the X chromosome. It affects approximately **8 percent of males** and only about **0.4 percent of females**. This disparity is due to the fact that the genes responsible for red-green color blindness are situated on the X chromosome. Males possess a **single X chromosome, whereas females have two**. The **Ishihara test** is a vision test used to detect colour blindness, especially **red-green colour deficiencies**.

BCW Fact:

- **Lymphatic filariasis (elephantiasis)** is a disfiguring disease caused by filarial worms — **Wuchereria bancrofti** (90% cases), **Brugia malayi**, and **Brugia timori**. It is transmitted by female mosquitoes (**Culex, Anopheles, and Aedes**) and mainly affects the lower limbs.
- **Taeniasis** is an intestinal parasitic infection caused by tapeworms — **Taenia solium** (pork tapeworm), **Taenia saginata** (beef tapeworm), and **Taenia asiatica**. **Taenia solium** is the most dangerous; its larvae can spread to organs, causing **neurocysticercosis (NCC)** when cysts form in the brain.
- **Oncogenes** are genes associated with cancer, and the study of cancer is called **Oncology**. German biologist Theodor Boveri predicted the existence of oncogenes in his 1914 book *Zur Frage der Entstehung Maligner Tumoren* (**Concerning the origin of Malignant Tumors**).
- **Cobalt-60** is a radioactive isotope used in **radiation therapy** for cancer treatment, especially **leukaemia**. It emits **gamma radiation** and is also used for sterilising medical equipment and industrial radiography.
- **Iodine-131**, an important radioisotope of iodine, was discovered by Glenn Seaborg and John Livingood in 1938. It is used to treat **thyroid cancer** and **hyperthyroidism**. It accumulates in thyroid cells, where radiation destroys affected cells with minimal impact on the rest of the body.